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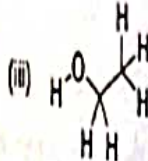
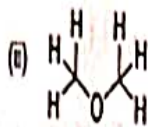
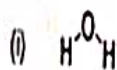
(A University under section 3 of the UGC Act, 1956)

End Semester Examinations**JULY 2023**Course Code: **CHY101**Course: **ENGINEERING CHEMISTRY**QP No. : **UF018-2**Duration: **3 hours**Max. Marks: **100****PART- A****Answer all the questions****10 x 2 = 20 Marks**

1. **Statement 1:** CaSO_4 deposit at the colder portion of the boiler.
Statement 2: Solubility of CaSO_4 decreases with a decrease in the temperature of the boiler.
(a) Statement 1 is correct; Statement 2 is wrong
(b) Statement 1 is wrong; Statement 2 is correct
(c) Statement 1 is correct; Statement 2 is correct
(d) Statement 1 is wrong; Statement 2 is wrong.
2. A given water sample contains 292 mg/L of magnesium bicarbonate, 648 mg/L of calcium bicarbonate and 100 mg/L of sodium chloride. Calculate the amount of hardness present in the water sample. (Atomic weight of Ca = 40; Mg = 24; H = 1; C = 12; O = 16; Na = 23; Cl = 35.45)
3. In the paper-making industry, the presence of minuscule quantities of silica and oil in the water cause deterioration of the quality of the paper. Suggest a suitable method to eliminate the trace quantities of silica and oil from the water.
4. Two identical iron nails are dipped separately in zeolite treated water and ion exchange resin treated water. Comment on the rate of corrosion in both nails.
5. The standard reduction potential of Al^{3+} is -1.66 V, whilst for Fe^{2+} is -0.44 V. Comment on the rate of complete oxidation of 1 kg of aluminium and iron in a moist environment.
6. Both polyethylene and Teflon have similar carbon backbone chain. However, the heat and corrosion resistance of Teflon is several folds higher

than that of polyethylene. Critically analyse the structure of both molecules and identify the reason behind the exceptional stability of Teflon.

7. Arrange the following molecules in the *increasing order* of their boiling point. Give a suitable explanation for your arrangement



8. Comment on the "aspect ratio" of one-dimensional nanomaterials.
9. Give any two examples of naturally available composite materials. Analyse their structure and name the components in them.
10. A commercial paint sample has the following problems (i) It takes a prolonged period of time for drying (ii) After drying, the paint surface produces numerous cracks. Identify the suitable chemicals that need to be added to the paint to overcome the aforementioned problems.

PART-B

Answer all the questions

4 x 15 = 60 Marks

11. (a) 100 mL hard water sample required 15 mL of 0.02 M EDTA solution. Calculate the hardness of water. Presume, Y liters of the same water sample was treated with zeolite softener. After treatment, the softener required 150 L of NaCl with 200 g/L for regeneration. Find out the total volume of water (Y). Also, predict whether this treated water can be utilized in high pressure boilers to generate steam. (8)
- (b) Industrial effluent enriched with pathogenic bacteria inadvertently mixed with fresh water pond. You are advised to utilize bleaching powder for disinfection. How will you evaluate the precise amount of bleaching powder necessary for disinfection? Explain with a suitable diagram to identify the required amount of bleaching powder for disinfection. Also, suggest any two methods to disinfect the water without increasing the hardness of the water. (7)

(OR)

12. (a) Ultra-Clean industry daily utilizes 10,000 L of hydrochloric acid enriched water for washing away fats and oils from the surface of the machinery. You are provided with 10,000 L of water with 2.34 g/L of calcium chloride and 3.12 g/L of magnesium chloride. How will you transform this hard water into hydrochloric acid enriched water through

an external treatment process? Explain the chosen external treatment process with a neat diagram and equations. Also, state whether hydrochloric acid enriched water is considered as soft water or hard water. Justify your answer. (8)

- (b) The quantum of potable water available on the earth is limited. Distillation is one of the energy-intensive processes for transforming brackish water into potable water. Can you suggest an energy-efficient alternative method that utilizes electricity to transform brackish water into clean water? Explain the chosen method in detail. (7)

13. (a) The national green hydrogen mission envisages the production of 5 million metric tonnes (MMT) of green hydrogen per annum by 2030. Devise an environmentally benign and efficient protocol for generating electricity by utilizing hydrogen gas. Delineate the chosen method with a suitable diagram and equations. (8)

- (b) TISCO Pvt Ltd laid LPG transmission iron pipelines in the eastern coastal region to seamlessly transmit petroleum products. Due to the region's high salinity and moisture content, the pipelines are getting easily corroded. Develop a suitable method to protect the pipelines from corrosion by utilizing electricity. Delineate the mechanism of the action of the chosen method. (7)

(OR)

14. (a) Electric vehicles are considered as the future means of transportation. They are used to curtail global warming by preventing greenhouse gas emissions. Choose a suitable rechargeable battery that can be utilized for electric vehicles. Analyse the key components in the chosen battery in detail and discuss the electrochemical reactions of the battery. (7)
- (b) Inorganic coatings provide weather and abrasion resistance to structures and enhance aesthetic appearances/toughness. Appraise all the available types of inorganic coatings in detail and mention the advantages of each type of coating. (8)

15. (a) Rockets and satellites involved in the deep-space exploration mission operate under extreme temperatures. Hence, lubricants used in such instruments should not lose their characteristic properties even under such extreme conditions. Suggest suitable methods to evaluate the highest operating temperature of the lubricants as well as to measure the viscosity index of the lubricant at variable temperatures. Delineate the chosen methods in detail. (12)

- (b) Both sodium chloride and magnesium oxide are ionic compounds. However, the melting point of the magnesium oxide is several folds higher than that of sodium chloride (MgO (2852°C) vs NaCl (801°C)). Analyze the reason behind this anomaly. (3)

(OR)

16. (a) Distinguish between the mechanism of hydrodynamic lubrication and boundary film lubrication in detail with appropriate examples. (10)
(b) Glycerol is a readily available non-toxic trihydric alcohol. With suitable equations, explain the conversion of glycerol into different types of semi-solid lubricants. Also, identify the kind of semi-solid lubricant utilized in (i) Ball bearings and (ii) Water pumps. (5)
17. (a) Metal nanoparticles are sub-micron scale entities made up of pure metals. Analyse the application of such noble metal nanoparticles (i) in the purification of water (ii) as anti-cancer agents. (7)
(b) With necessary diagrams, distinguish between physical vapour deposition and chemical vapour deposition in detail (8)
- (OR)
18. (a) Appraise different types of particle-reinforced composite materials and fiber-reinforced composite materials in detail. Delineate their applications. (10)
(b) Evaluate the applications of metal oxide nanoparticles in the electronic industries and in the making of self-cleaning glasses. (5)

PART-C

Answer the following

1 x 20 = 20 Marks

19. (a) An iron boiler is filled with oxygen-enriched alkaline water. Write down the anodic and cathodic reactions of iron with the alkaline water. Analyse the factors that influence electrochemical corrosion in detail. Identify suitable chemicals to remove dissolved oxygen and other dissolved gases from the boiler water. With suitable equations, explain the mode of action of the chosen chemicals. (10)
(b) You are provided with 1 kg of graphite block and asked to convert it to the corresponding graphite nanopowder by utilizing the ball mill method. Evaluate all the necessary parameters in the ball milling method which affect the product particle size. Appraise the structural significance of the graphite and critically analyze its role as a solid lubricant. (10)

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